

Molecular mechanisms of alcohol-mediated carcinogenesis.

Approximately 3.6% of cancers worldwide derive from chronic alcohol drinking, including those of the upper aerodigestive tract, the liver, the colorectum and the breast. Although the mechanisms for alcohol-associated carcinogenesis are not completely understood, most recent research has focused on acetaldehyde, the first and most toxic ethanol metabolite, as a cancer-causing agent. Ethanol may also stimulate carcinogenesis by inhibiting DNA methylation and by interacting with retinoid metabolism. Alcohol-related carcinogenesis may interact with other factors such as smoking, diet and comorbidities, and depends on genetic susceptibility.